CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

JOHN & NICOLE SANTOS DAIRY #2

5245 S HIGHWAY 33

GUSTINE CA 95322

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John & Nicole Santos Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
5245 S HWY 33	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1942

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X240-X003-XXXX

B. OPERATORS

Duarte, Joe

Operator name: Duarte, Joe

Telephone no.: (209) 993-8662 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5245 SE Hwy 33	Gustine	CA	95322

Santos, John & Nicole

Operator name: Santos, John & Nicole

Telephone no.: (209) 968-7832 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
22643 S Mercy Springs RD	Los Banos	CA	93635

This operator is responsible for paying permit fees.

C. OWNERS

Duarte, Joe

Legal owner name: Duarte, Joe

Telephone no.: (209) 993-8662 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5245 SE Hwy 33	Gustine	CA	95322

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	510	0	0	0	0	0
Maximum number	510	0	0	0	0	0
Average number	510	0	0	0	0	0
Avg live weight (lbs)	1,400	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 13,262.92 *tons per reporting period*

Total nitrogen from manure: 173,391.57 *lbs per reporting period*

After ammonia losses (30% loss applied): 121,374.10 *lbs per reporting period*

Total phosphorus from manure: 29,388.38 *lbs per reporting period*

Total potassium from manure: 98,674.92 *lbs per reporting period*

Total salt from manure: 240,133.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 42,120,000 *gallons*

Total nitrogen generated: 102,834.79 *lbs*

Total phosphorus generated: 21,498.42 *lbs*

Total potassium generated: 125,856.15 *lbs*

Total salt generated: 937,658.89 *lbs*

	42,120,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	42,120,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	30	30	2	manure	X063-X240-X003-XXXX
2	54	54	2	process wastewater	X063-X240-X003-XXXX
3	68	68	2	process wastewater	X063-X240-X003-XXXX
4	53	53	2	process wastewater	X063-X240-X003-XXXX
Totals for areas that were used for application	205	205	8		
Totals for areas that were not used for application					
Land application area totals	205	205	8		

B. CROPS AND HARVESTS

1

Field name: 1

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	600.00 *ton*	Dry-weight		70.5	14,700.00	2,300.00	22,700.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.00	173.46	27.14	267.86	1,109.20

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	840.00 *ton*	Dry-weight		64.0	12,500.00	2,300.00	9,400.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.00	252.00	46.37	189.50	1,189.44

2

Field name: 2

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 54 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,150.00 *ton*	Dry-weight		66.7	15,400.00	2,700.00	22,000.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	21.30	218.42	38.30	312.03	1,134.67

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,320.00 *ton*	Dry-weight		67.2	16,200.00	3,100.00	20,000.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.44	259.78	49.71	320.71	1,250.77

3

Field name: 3

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 68 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,100.00 *ton*	Dry-weight		74.8	16,200.00	2,700.00	19,900.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.18	132.08	22.01	162.24	741.92

3

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	2,098.00 *ton*	Dry-weight		59.7	14,600.00	2,400.00	13,800.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.85	363.07	59.68	343.17	1,591.52

4

Field name: 4

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	1,442.00 *ton*	Dry-weight		75.2	21,200.00	3,400.00	27,600.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	27.21	286.09	45.88	372.46	1,309.01

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	1,430.00 *ton*	Dry-weight		73.8	15,000.00	3,000.00	21,400.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.98	212.07	42.41	302.56	1,159.33

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/22/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough — Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	216.75	96.67	297.14	0.00	480.00 *ton*
Application event totals		216.75	96.67	297.14	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.51	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		7.51	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.01	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		6.01	0.00	0.00	0.00	

1 - 05/01/2023: Corn, silage

Field name: 1

Crop: Corn, silage — Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	311.26	43.96	45.14	0.00	675.00 *ton*
Application event totals		311.26	43.96	45.14	0.00	

1 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.49	6,192,100.00 *gal*
Application event totals		1.21	0.00	0.00	344.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.81	0.00	0.00	0.00	5,616,000.00 *gal*
Canal	Surface water	0.53	0.00	0.00	152.30	2,737,560.00 *gal*
Application event totals		8.34	0.00	0.00	152.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.27	0.00	0.00	76.15	1,368,780.00 *gal*
Application event totals		0.27	0.00	0.00	76.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.71	0.00	0.00	0.00	6,984,000.00 *gal*
Canal	Surface water	0.38	0.00	0.00	108.79	1,955,400.00 *gal*
Application event totals		10.09	0.00	0.00	108.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.30	0.00	0.00	87.03	1,564,320.00 *gal*
Application event totals		0.30	0.00	0.00	87.03	

1 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.21	0.00	0.00	0.00	6,624,000.00 *gal*
Canal	Surface water	0.41	0.00	0.00	116.04	2,085,760.00 *gal*
Application event totals		9.62	0.00	0.00	116.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.44	0.00	0.00	126.92	2,281,300.00 *gal*
Application event totals		0.44	0.00	0.00	126.92	

2 - 10/22/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	82.57	21.73	99.64	951.33	3,240,000.00 *gal*
Irrigation Well	Ground water	6.68	0.00	0.00	0.00	8,640,000.00 *gal*
Application event totals		89.24	21.73	99.64	951.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.68	20.43	114.76	811.13	1,296,000.00 *gal*
Application event totals		97.68	20.43	114.76	811.13	

2 - 10/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.01	19.91	123.37	878.56	2,016,000.00 *gal*
Irrigation Well	Ground water	6.40	0.00	0.00	0.00	8,280,000.00 *gal*
Application event totals		106.40	19.91	123.37	878.56	

2 - 05/20/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.58	15.64	96.94	690.30	1,584,000.00 *gal*
Canal	Surface water	1.36	0.00	0.00	389.81	12,612,330.00 *gal*
Application event totals		79.94	15.64	96.94	1,080.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	82.15	16.35	101.34	721.68	1,656,000.00 *gal*
Irrigation Well	Ground water	6.68	0.00	0.00	0.00	8,640,000.00 *gal*
Canal	Surface water	0.60	0.00	0.00	172.24	5,572,890.00 *gal*
Application event totals		89.43	16.35	101.34	893.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.30	0.00	0.00	86.63	2,802,740.00 *gal*
Application event totals		0.30	0.00	0.00	86.63	

2 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.72	17.06	105.75	753.05	1,728,000.00 *gal*
Irrigation Well	Ground water	6.95	0.00	0.00	0.00	9,000,000.00 *gal*
Canal	Surface water	0.44	0.00	0.00	125.91	4,073,750.00 *gal*
Application event totals		93.11	17.06	105.75	878.96	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.32	0.00	0.00	90.65	2,933,100.00 *gal*
Application event totals		0.32	0.00	0.00	90.65	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.72	17.06	105.75	753.05	1,728,000.00 *gal*
Irrigation Well	Ground water	7.23	0.00	0.00	0.00	9,360,000.00 *gal*
Canal	Surface water	0.42	0.00	0.00	120.87	3,910,800.00 *gal*
Application event totals		93.38	17.06	105.75	873.93	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.49	0.00	0.00	139.00	4,497,420.00 *gal*
Application event totals		0.49	0.00	0.00	139.00	

3 - 10/22/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 10/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.57	17.26	79.13	755.47	3,240,000.00 *gal*
Irrigation Well	Ground water	5.96	0.00	0.00	0.00	9,720,000.00 *gal*
Application event totals		71.53	17.26	79.13	755.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.27	19.76	122.47	872.10	2,520,000.00 *gal*
Irrigation Well	Ground water	5.74	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		105.01	19.76	122.47	872.10	

3 - 05/20/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.78	21.46	132.96	946.85	2,736,000.00 *gal*
Canal	Surface water	1.54	0.00	0.00	440.74	17,957,090.00 *gal*
Application event totals		109.32	21.46	132.96	1,387.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.11	20.33	125.96	897.02	2,592,000.00 *gal*
Irrigation Well	Ground water	6.19	0.00	0.00	0.00	10,080,000.00 *gal*
Canal	Surface water	0.67	0.00	0.00	191.17	7,789,010.00 *gal*
Application event totals		108.96	20.33	125.96	1,088.19	

3 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.34	0.00	0.00	96.79	3,943,390.00 *gal*
Application event totals		0.34	0.00	0.00	96.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.94	20.89	129.46	921.94	2,664,000.00 *gal*
Irrigation Well	Ground water	5.52	0.00	0.00	0.00	9,000,000.00 *gal*
Canal	Surface water	0.49	0.00	0.00	139.18	5,670,660.00 *gal*
Application event totals		110.95	20.89	129.46	1,061.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.37	0.00	0.00	104.79	4,269,290.00 *gal*
Application event totals		0.37	0.00	0.00	104.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.94	20.89	129.46	921.94	2,664,000.00 *gal*
Irrigation Well	Ground water	6.10	0.00	0.00	0.00	9,936,000.00 *gal*
Canal	Surface water	0.49	0.00	0.00	140.78	5,735,840.00 *gal*
Application event totals		111.53	20.89	129.46	1,062.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.55	0.00	0.00	158.38	6,452,820.00 *gal*
Application event totals		0.55	0.00	0.00	158.38	

4 - 10/22/2022: Oats, silage-soft dough

4 - 10/22/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.82	27.06	124.08	1,184.68	3,960,000.00 *gal*
Irrigation Well	Ground water	4.87	0.00	0.00	0.00	6,192,000.00 *gal*
Application event totals		107.69	27.06	124.08	1,184.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.52	20.81	116.93	826.44	1,296,000.00 *gal*
Application event totals		99.52	20.81	116.93	826.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.89	20.28	125.70	895.14	2,016,000.00 *gal*
Irrigation Well	Ground water	4.25	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		106.14	20.28	125.70	895.14	

4 - 05/01/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.22	11.59	71.83	511.51	1,152,000.00 *gal*
Canal	Surface water	1.31	0.00	0.00	374.59	11,895,350.00 *gal*
Application event totals		59.54	11.59	71.83	886.10	

4 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.78	14.49	89.79	639.38	1,440,000.00 *gal*
Irrigation Well	Ground water	6.68	0.00	0.00	0.00	8,486,000.00 *gal*
Canal	Surface water	0.59	0.00	0.00	167.28	5,312,170.00 *gal*
Application event totals		80.05	14.49	89.79	806.67	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	83.13	2,639,790.00 *gal*
Application event totals		0.29	0.00	0.00	83.13	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.22	11.59	71.83	511.51	1,152,000.00 *gal*
Irrigation Well	Ground water	7.99	0.00	0.00	0.00	10,152,000.00 *gal*
Canal	Surface water	0.42	0.00	0.00	119.05	3,780,440.00 *gal*
Application event totals		66.63	11.59	71.83	630.56	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.32	0.00	0.00	92.36	2,933,100.00 *gal*
Application event totals		0.32	0.00	0.00	92.36	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.78	14.49	89.79	639.38	1,440,000.00 *gal*
Irrigation Well	Ground water	7.88	0.00	0.00	0.00	10,008,000.00 *gal*
Canal	Surface water	0.43	0.00	0.00	123.15	3,910,800.00 *gal*
Application event totals		81.09	14.49	89.79	762.54	

4 - 05/01/2023: Corn, silage

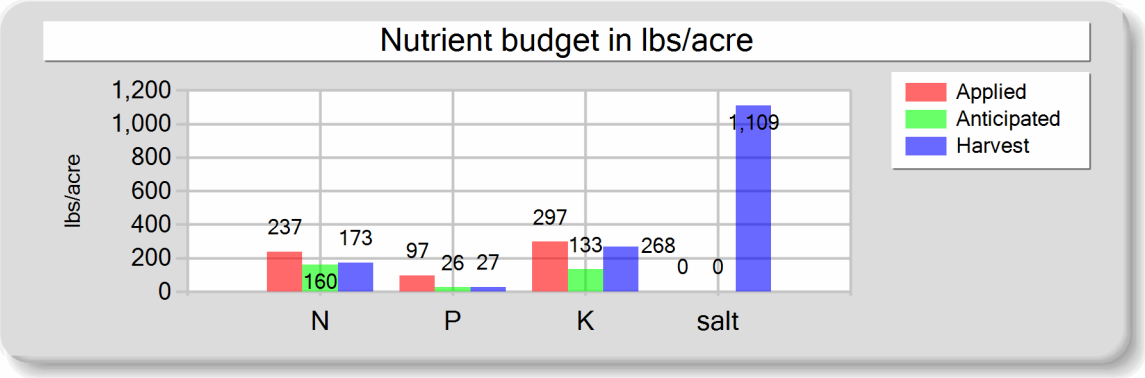
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.48	0.00	0.00	137.52	4,367,060.00 *gal*
Application event totals		0.48	0.00	0.00	137.52	

B. NUTRIENT BUDGET

1 - 10/22/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	216.75	96.67	297.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.27	96.67	297.14	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.46	27.14	267.86	1,109.20
Nutrient balance	63.81	69.53	29.28	-1,109.20
Applied to removed ratio	1.37	3.56	1.11	0.00

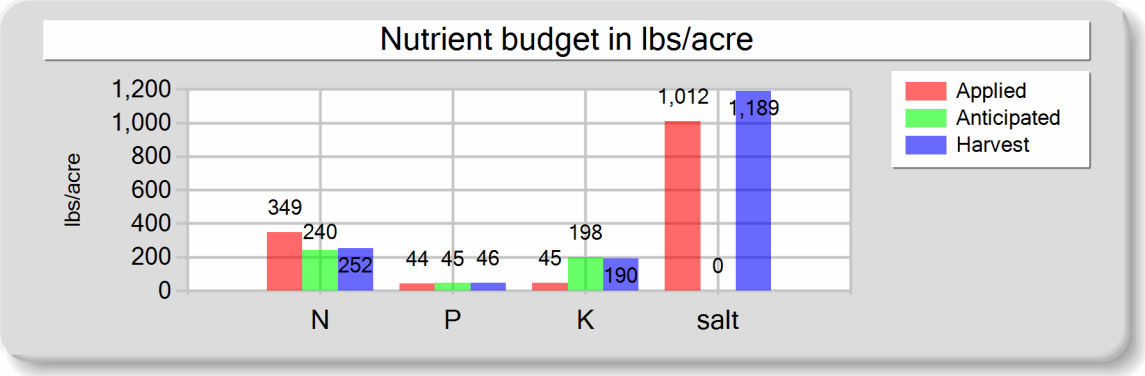
Fresh water applied	
9,720,000.00	*gallons*
357.95	*acre-inches*
11.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/01/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	311.26	43.96	45.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	30.28	0.00	0.00	1,011.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	348.53	43.96	45.14	1,011.70
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.00	46.37	189.50	1,189.44
Nutrient balance	96.53	-2.41	-144.36	-177.74
Applied to removed ratio	1.38	0.95	0.24	0.85

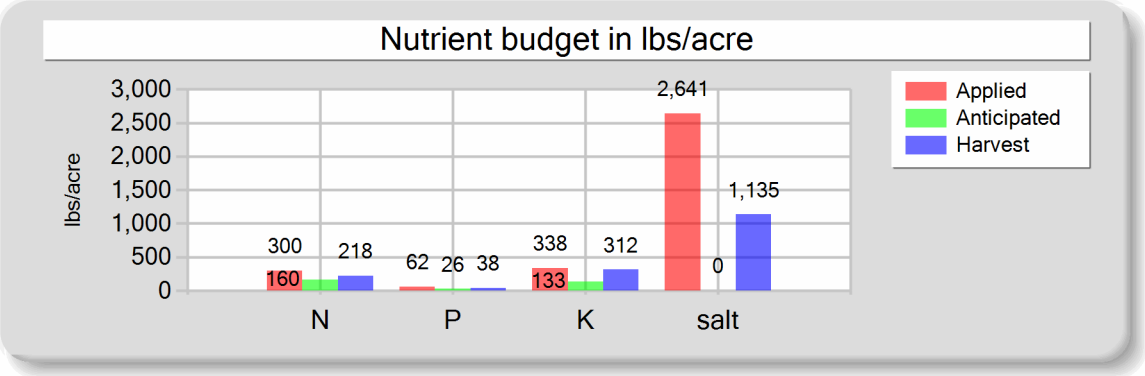
Fresh water applied	
37,409,220.00	*gallons*
1,377.65	*acre-inches*
45.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/22/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.25	62.07	337.77	2,641.03
Fresh water	13.07	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.32	62.07	337.77	2,641.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	218.42	38.30	312.03	1,134.67
Nutrient balance	81.90	23.77	25.74	1,506.36
Applied to removed ratio	1.37	1.62	1.08	2.33

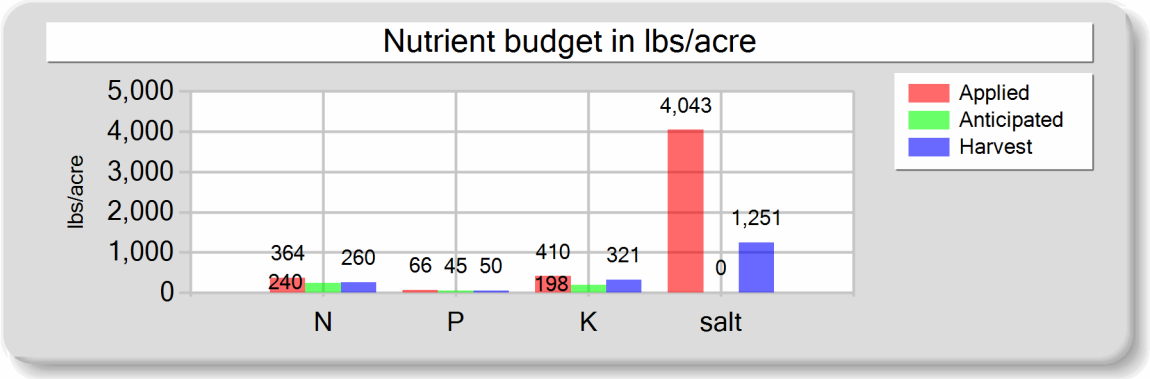
Fresh water applied	
16,920,000.00	*gallons*
623.11	*acre-inches*
11.54	*inches/acre*

Process wastewater applied	
6,552,000.00	*gallons*
241.29	*acre-inches*
4.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/20/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	332.16	66.12	409.77	2,918.08
Fresh water	24.80	0.00	0.00	1,125.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	363.96	66.12	409.77	4,043.20
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	259.78	49.71	320.71	1,250.77
Nutrient balance	104.19	16.41	89.06	2,792.43
Applied to removed ratio	1.40	1.33	1.28	3.23

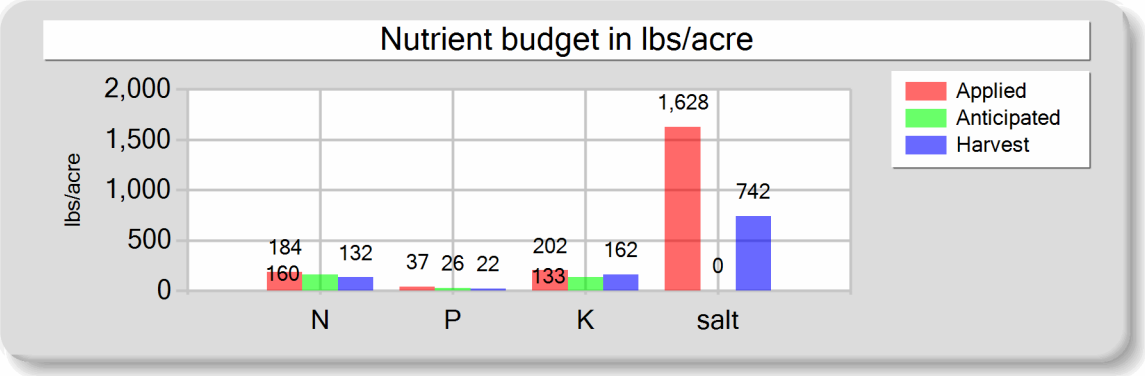
Fresh water applied	
63,403,030.00	*gallons*
2,334.92	*acre-inches*
43.24	*inches/acre*

Process wastewater applied	
6,696,000.00	*gallons*
246.59	*acre-inches*
4.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/22/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.84	37.02	201.59	1,627.57
Fresh water	11.71	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.55	37.02	201.59	1,627.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.08	22.01	162.24	741.92
Nutrient balance	51.47	15.00	39.35	885.65
Applied to removed ratio	1.39	1.68	1.24	2.19

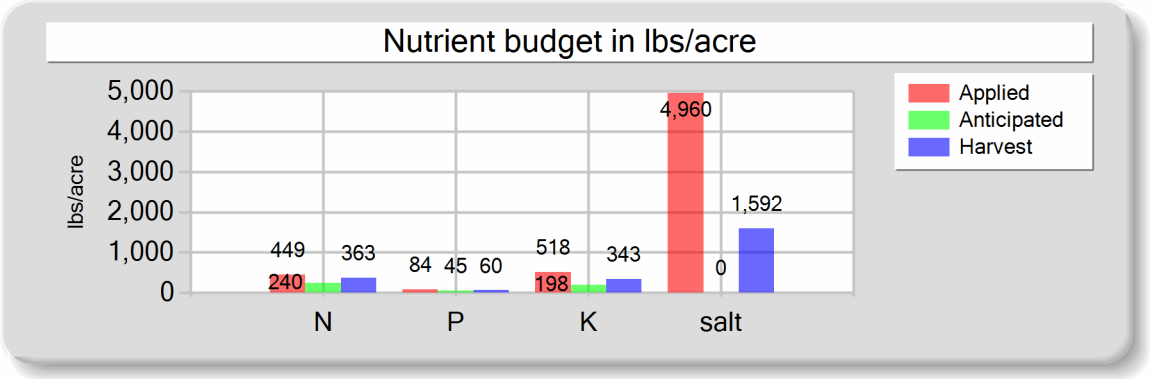
Fresh water applied	
19,080,000.00	*gallons*
702.65	*acre-inches*
10.33	*inches/acre*

Process wastewater applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
3.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/20/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	419.78	83.56	517.85	3,687.74
Fresh water	22.26	0.00	0.00	1,271.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	449.03	83.56	517.85	4,959.57
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	363.07	59.68	343.17	1,591.52
Nutrient balance	85.97	23.88	174.68	3,368.06
Applied to removed ratio	1.24	1.40	1.51	3.12

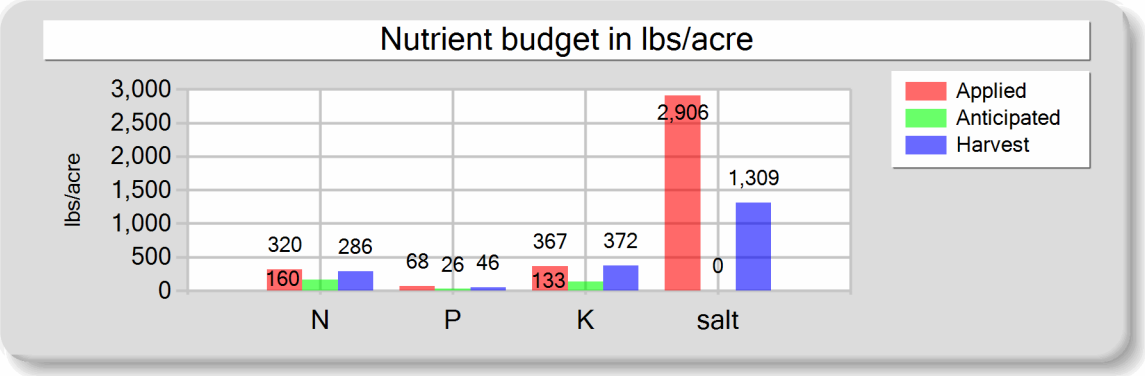
Fresh water applied	
80,834,100.00	*gallons*
2,976.85	*acre-inches*
43.78	*inches/acre*

Process wastewater applied	
10,656,000.00	*gallons*
392.42	*acre-inches*
5.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/22/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	304.23	68.16	366.71	2,906.25
Fresh water	9.13	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.36	68.16	366.71	2,906.25
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	286.09	45.88	372.46	1,309.01
Nutrient balance	34.26	22.28	-5.76	1,597.24
Applied to removed ratio	1.12	1.49	0.98	2.22

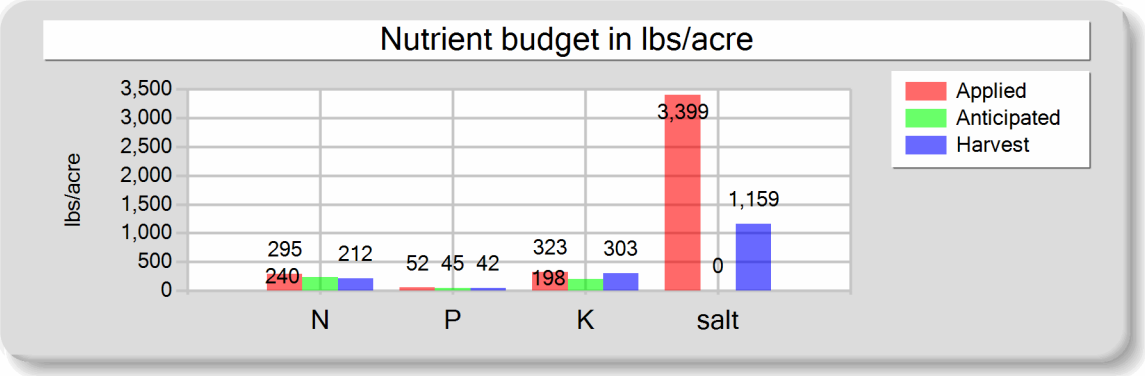
Fresh water applied	
11,592,000.00	*gallons*
426.89	*acre-inches*
8.05	*inches/acre*

Process wastewater applied	
7,272,000.00	*gallons*
267.80	*acre-inches*
5.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/01/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.01	52.16	323.23	2,301.78
Fresh water	26.39	0.00	0.00	1,097.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.40	52.16	323.23	3,398.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.07	42.41	302.56	1,159.33
Nutrient balance	83.33	9.74	20.67	2,239.55
Applied to removed ratio	1.39	1.23	1.07	2.93

Fresh water applied	
63,484,710.00	*gallons*
2,337.93	*acre-inches*
44.11	*inches/acre*

Process wastewater applied	
5,184,000.00	*gallons*
190.91	*acre-inches*
3.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Corral Manure

Sample and source description: 2022 Fall Corral Manure

Sample date: 10/05/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,300.00	9,500.00	29,200.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 09/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,700.00	8,700.00	25,700.00							0.00
DL	0.01	0.01	0.01							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 12/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	3,700.00	3,800.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	164.00	144.00	0.00	0.90	43.40	199.00								3,940.00	1,900
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	486.00	305.00	0.00	1.70	102.00	573.00								7,110.00	4,050
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	320.00	260.00	0.00	1.00	63.90	396.00								6,060.00	2,820
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	776.00	370.00	0.00	0.30	40.40	148.00								2,640.00	1,310
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Slurry

Sample and source description: Slurry

Sample date: 10/30/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	653.00	216.00	0.00	1.20	87.10	316.00								3,730.00	1,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								328.00	200
DL	0.01	0.01	0.01								10.00	1

Domestic Well

Domestic Well Milk Barn

Sample description: Domestic Well Milk Barn

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.80	0.00	4.80								1,080.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

IW

Sample description: IW

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.00	0.00	5.00								1,110.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/22/2022: Oats, silage-soft dough

Field 1 Oats

Sample and source description: Field 1 Oats

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	2,300.00	22,700.00		9.40
DL	0.01	0.01	0.01		1.00

1 - 05/01/2023: Corn, silage

1 - 05/01/2023: Corn, silage

Field 1 Corn

Sample and source description: Field 1 Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,300.00	9,400.00		5.90
DL	0.01	0.01	0.01		1.00

2 - 10/22/2022: Oats, silage-soft dough

Field 2 Oats

Sample and source description: Field 2 Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,700.00	22,000.00		8.00
DL	0.01	0.01	0.01		1.00

2 - 05/20/2023: Corn, silage

Field 2 Corn

Sample and source description: Field 2 Corn

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,100.00	20,000.00		7.80
DL	0.01	0.01	0.01		1.00

3 - 10/22/2022: Oats, silage-soft dough

Field 3 Oats

Sample and source description: Field 3 Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	2,700.00	19,900.00		9.10
DL	0.01	0.01	0.01		1.00

3 - 05/20/2023: Corn, silage

Field 3 Corn

Sample and source description: Field 3 Corn

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,400.00	13,800.00		6.40
DL	0.01	0.01	0.01		1.00

4 - 10/22/2022: Oats, silage-soft dough

Field 4 Oats

Sample and source description: Field 4 Oats

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	3,400.00	27,600.00		9.70
DL	0.01	0.01	0.01		1.00

4 - 05/01/2023: Corn, silage

Field 4 Corn

Sample and source description: Field 4 Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	3,000.00	21,400.00		8.20
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

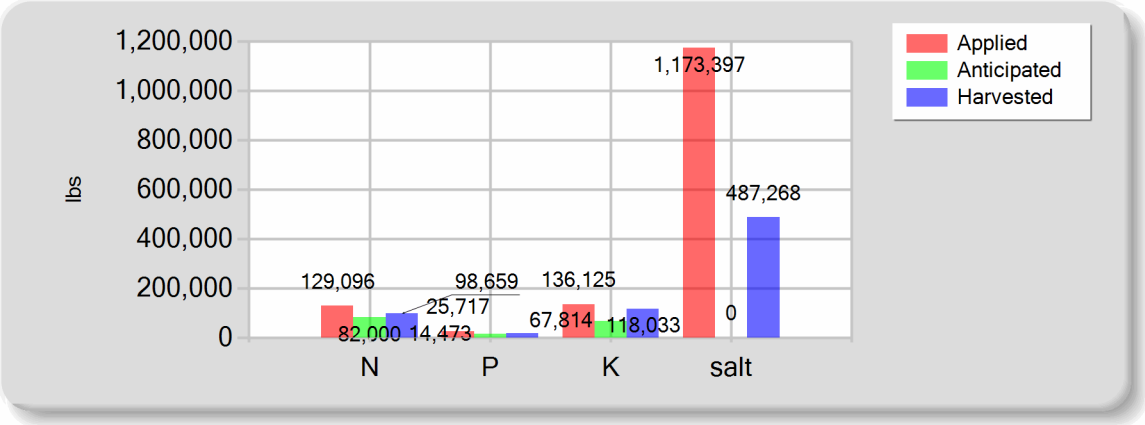
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

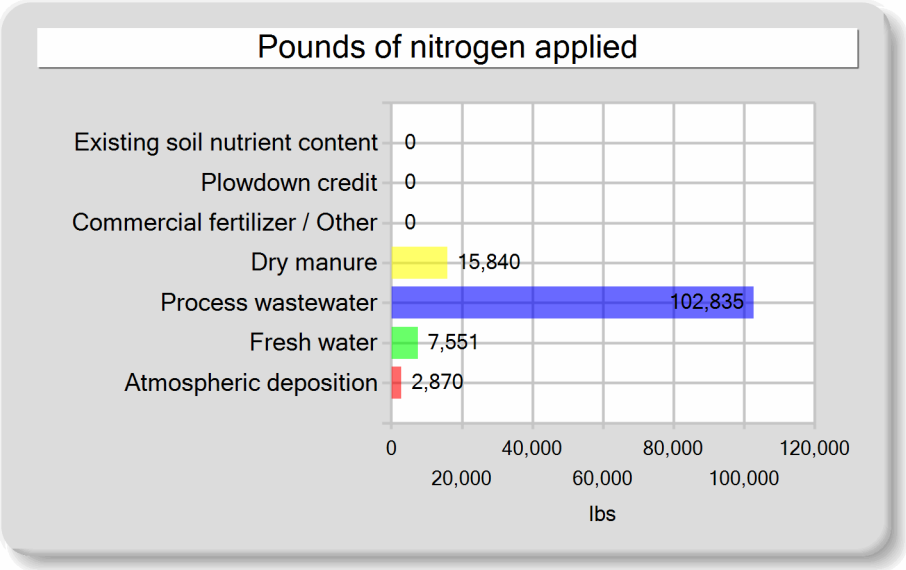
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

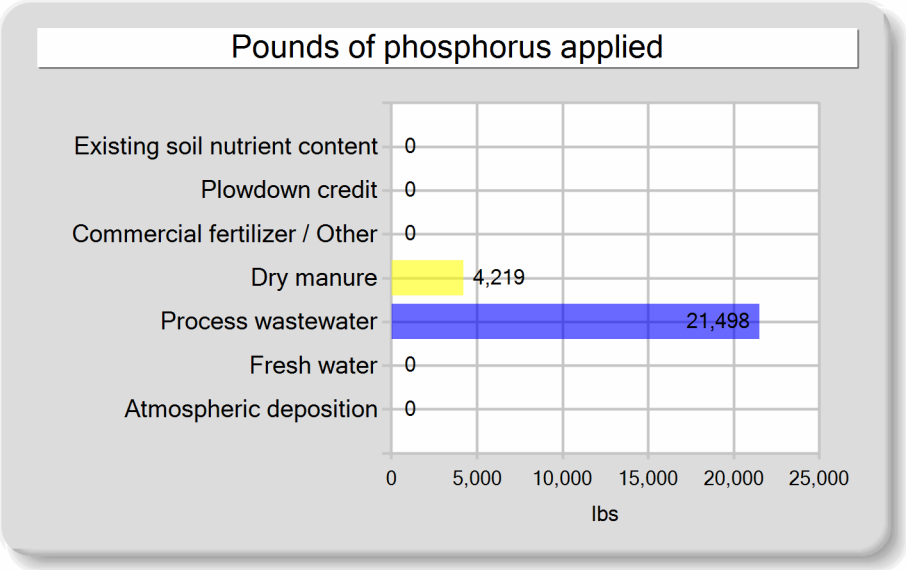
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,840.14	4,218.84	10,268.50	0.00
Process wastewater	102,834.79	21,498.42	125,856.15	937,658.89
Fresh water	7,551.07	0.00	0.00	235,738.01
Atmospheric deposition	2,870.00	0.00	0.00	0.00
Total nutrients applied	129,096.00	25,717.26	136,124.65	1,173,396.89
Anticipated crop nutrient removal	82,000.00	14,473.00	67,814.00	0.00
Actual crop nutrient removal	98,658.99	17,192.52	118,033.17	487,268.34
Nutrient balance	30,437.01	8,524.73	18,091.48	686,128.56
Applied to removed ratio	1.31	1.50	1.15	2.41

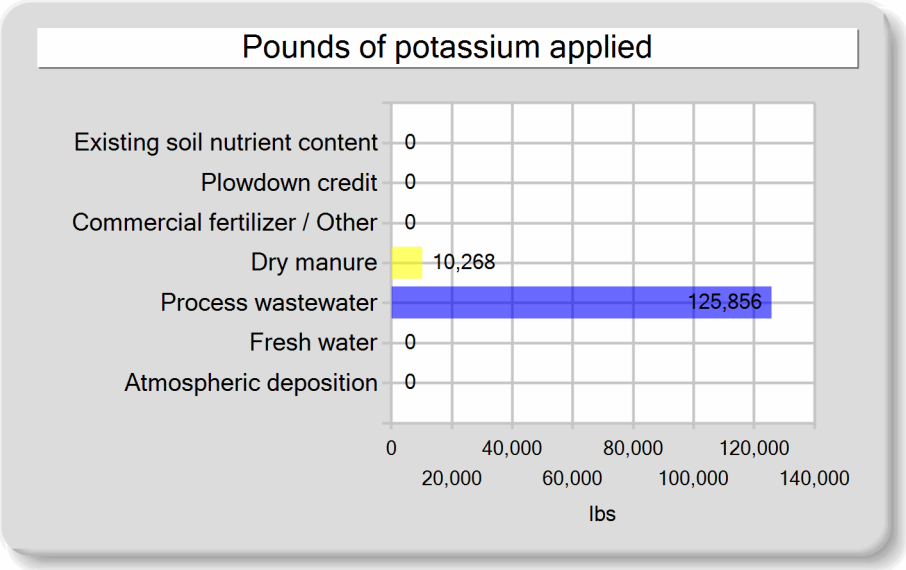
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

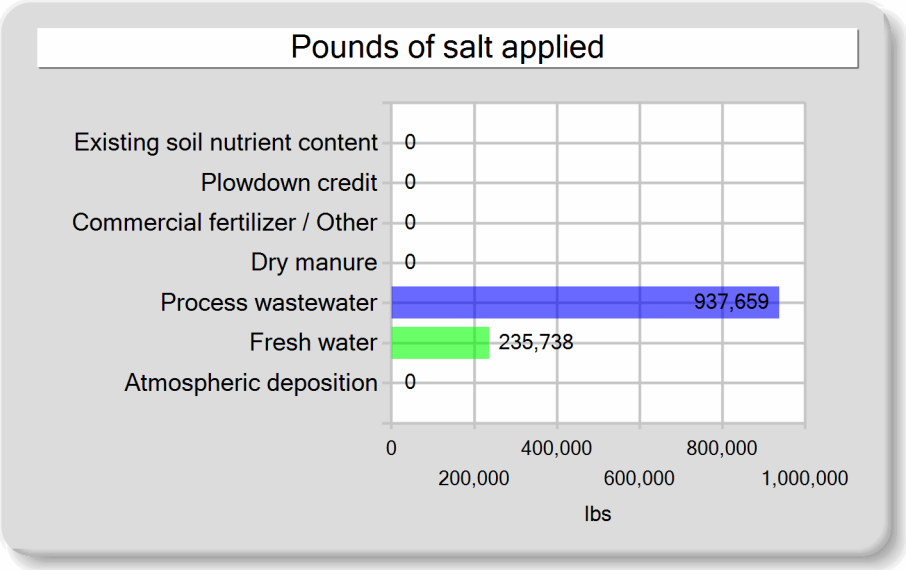


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe Duarte	John & Nicole Santos
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/28/24	Jun 28, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23G2094
Sampled By: M.Pedroso
Requested By: Joe Duarte
Submitted Date: 07/27/23
Reported Date: 08/03/23
Project: John & Nicole Santos #2
Crop ID:

Joe Duarte
5245 S Hwy 33
Gustine CA 95322
00-0022618 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	IW	7/26/23	9:28	1110	1.11	5	*		7.6

***** = Field NH_4-N not Taken.

ND = None Detected

Approved By: Scott M Friedland

Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23G2094
Sampled By: M.Pedroso
Requested By: Joe Duarte
Submitted Date: 07/27/23
Reported Date: 08/03/23
Project: John & Nicole Santos #2
Crop ID:

Joe Duarte
5245 S Hwy 33
Gustine CA 95322
00-0022618 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	IW	7/26/23	9:28	**1110**	**1.11**	**5**	13.65	1.14

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Joe Duarte
5245 S Hwy 33
Gustine, CA 95322

Account# 00-0022618
Account Manager: Ben Nydam
Submitted By: Joe Duarte
Ranch: John & Nicole Santos #2

Received: 07/27/2023 16:00
Reported: 08/03/2023 13:16

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23G2094-01	IW	Waste Water	M.Pedroso		07/26/2023 9:28

Default Cooler Temperature on Receipt °C: 0.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Joe Duarte
5245 S Hwy 33
Gustine, CA 95322

Account# 00-0022618
Account Manager: Ben Nydam
Submitted By: Joe Duarte
Ranch: John & Nicole Santos #2

Received: 07/27/2023 16:00
Reported: 08/03/2023 13:16

Sample Results

Sample: IW
23G2094-01 (Water)

Sampled: 7/26/2023 9:28
Sampled By: M.Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.11**	mmhos/cm	0.01	1		07/31/23 15:16	SM 2510 B		BEG1123
Electrical Conductivity umhos	**1110**	umhos/cm	10.0	1		07/31/23 15:16	SM 2510 B		BEG1123
Ammonia (as N)	*	mg/L	0.00	1		07/26/23 09:28	Field		BEG1095
Nitrate Nitrogen as NO3N	**5.0**	mg/L	0.1	1	10	07/27/23 22:26	EPA 300.0		BEG1056
pH	**7.6**	units	1.0	1		07/31/23 15:16	SM 4500-H+	H	BEG1123
Temperature	**25.0**	°C	0.0	1		07/31/23 15:16	SM 2510 B		BEG1123

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DELLAVALLE™
LABORATORY INC

Joe Duarte
5245 S Hwy 33
Gustine, CA 95322

Account# 00-0022618
Account Manager: Ben Nydam
Submitted By: Joe Duarte
Ranch: John & Nicole Santos #2

Received: 07/27/2023 16:00
Reported: 08/03/2023 13:16

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1056										
Blank (BEG1056-BLK1)					Prepared & Analyzed: 7/27/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG1056-BLK2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEG1056-BS1)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
Duplicate (BEG1056-DUP1)		**Source: 23G2094-01**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L		5.0			0.343	10
Matrix Spike (BEG1056-MS1)		**Source: 23G2094-01**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	10.5		0.1	mg/L	5.000	5.0	111	90-110		
Reference (BEG1056-SRM1)					Prepared & Analyzed: 7/27/2023					
Nitrate Nitrogen as NO3N	9.5			mg/L	10.00		95.4	90-110		
Reference (BEG1056-SRM2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		96.2	90-110		

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DELLAVALLE™ LABORATORY INC

Joe Duarte
5245 S Hwy 33
Gustine, CA 95322

Account# 00-0022618
Account Manager: Ben Nydam
Submitted By: Joe Duarte
Ranch: John & Nicole Santos #2

Received: 07/27/2023 16:00
Reported: 08/03/2023 13:16

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1123										
Blank (BEG1123-BLK1)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEG1123-BLK2)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEG1123-BLK3)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	5.6		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEG1123-DUP1)		**Source: 23G2091-02**			Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	0.26		0.01	mmhos/cm		0.26			1.08	10
pH	7.7		1.0	units		7.6			1.31	10
Electrical Conductivity umhos	258		10.0	umhos/cm		261			1.08	10
Duplicate (BEG1123-DUP2)		**Source: 23G2097-05**			Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	7.8		1.0	units		7.7			0.516	10
Electrical Conductivity	0.98		0.01	mmhos/cm		0.98			0.194	10
Electrical Conductivity umhos	980		10.0	umhos/cm		978			0.194	10
Reference (BEG1123-SRM1)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	508			umhos/cm	538.0		94.5	90-110		
Reference (BEG1123-SRM2)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	7.8			units	7.790		100	.7163-101.28		
Reference (BEG1123-SRM3)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	945			umhos/cm	1000		94.5	90-110		
Electrical Conductivity umhos	945			umhos/cm	1000		94.5	90-110		
Reference (BEG1123-SRM4)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	942			umhos/cm	1000		94.2	90-110		
Electrical Conductivity umhos	942			umhos/cm	1000		94.2	90-110		
Reference (BEG1123-SRM5)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	952			umhos/cm	1000		95.2	90-110		

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DELLAVALLE™
LABORATORY INC

Joe Duarte
5245 S Hwy 33
Gustine, CA 95322

Account# 00-0022618
Account Manager: Ben Nydam
Submitted By: Joe Duarte
Ranch: John & Nicole Santos #2

Received: 07/27/2023 16:00
Reported: 08/03/2023 13:16

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1123 (Continued)										
Reference (BEG1123-SRM5)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity umhos	952			umhos/cm	1000		95.2	90-110		
Reference (BEG1123-SRM6)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG1123-SRM7)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG1123-SRM8)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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07/27/23 16:00 23G2094

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 22618 Cons. 8

Purchase Order No. Results Needed By

Client: Duarte, Joe
Address: 5245 S Hwy 33
City, State, Zip: Gustine, CA 95322
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Joe Duarte

Ranch: Jonnt Nicole Santos #2

Date sampled: 7/26/23

Sampled by: M Pedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 1 No. Bottles:

Water Type: [ ] Drinking [X] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. IW	Sampled From:	7/26/23	9:28A		0.3
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Pedroso	CAS INC	7/26/23 9:28A	7/27/23 11Am
Second	JLF	DLV	7-27-23 11:00	7-27-23 3:35
Third				
Fourth		AW	7/27 16:00	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___

Shipping
$ ___ In
$ ___ Out
Date ___

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

07/27/23 16:00 23G2094

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos CA 93635
00-0022364 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0967
Sampled By: M.Dekeyrel
Requested By: John & Nicole Santos
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW-MB	12/13/23	11:30	1080	1.08	4.8	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos CA 93635
00-0022364 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0967
Sampled By: M.Dekeyrel
Requested By: John & Nicole Santos
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW-MB	12/13/23	11:30	**1080**	**1.08**	**4.8**	13.10	1.09

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos, CA 93635

Account# 00-0022364
Account Manager: Ben Nydam
Submitted By: John & Nicole Santos

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:14

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0967-01	DW-MB	Ag Water	M.Dekeyrel		12/13/2023 11:30

Default Cooler Temperature on Receipt °C: 4.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos, CA 93635

Account# 00-0022364
Account Manager: Ben Nydam
Submitted By: John & Nicole Santos

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:14

Sample Results

Sample: DW-MB
23L0967-01 (Water)

Sampled: 12/13/2023 11:30
Sampled By: M.Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		12/18/23 16:58	SM 2510 B		BEL0701
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		12/18/23 16:58	SM 2510 B		BEL0701
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 11:30	Field		BEL0648
Nitrate Nitrogen as NO3N	**4.8**	mg/L	0.1	1	10	12/15/23 13:26	EPA 300.0		BEL0595
Temperature	**25.0**	units	0.0	1		12/18/23 16:58	SM 4500-H+	H	BEL0701
pH	**7.6**	units	1.0	1		12/18/23 16:58	SM 4500-H+	H	BEL0701

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DELLAVALLE™ LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos, CA 93635

Account# 00-0022364
Account Manager: Ben Nydam
Submitted By: John & Nicole Santos

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:14

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0595										
Blank (BEL0595-BLK1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0595-BLK2)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0595-BLK3)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0595-BLK4)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0595-BLK5)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0595-BS1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.1	90-110		
LCS (BEL0595-BS2)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.9	90-110		
LCS (BEL0595-BS3)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0595-BS4)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
Duplicate (BEL0595-DUP1)		**Source: 23L0848-01**			Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.6		0.1	mg/L		0.6			0.630	10
Duplicate (BEL0595-DUP2)		**Source: 23L0877-02**			Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	0.08		0.1	mg/L		0.08			1.27	10
Duplicate (BEL0595-DUP3)		**Source: 23L0909-01**			Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	2.4		0.1	mg/L		2.4			0.371	10
Duplicate (BEL0595-DUP4)		**Source: 23L0963-01**			Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	19.5		0.1	mg/L		19.5			0.313	10
Matrix Spike (BEL0595-MS1)		**Source: 23L0848-01**			Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	5.9		0.1	mg/L	5.000	0.6	105	90-110		
Matrix Spike (BEL0595-MS2)		**Source: 23L0877-02**			Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.08	104	90-110		

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DELLAVALLE™ LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos, CA 93635

Account# 00-0022364
Account Manager: Ben Nydam
Submitted By: John & Nicole Santos

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0595 (Continued)										
Matrix Spike (BEL0595-MS3)			**Source: 23L0909-01**		Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	7.7		0.1	mg/L	5.000	2.4	105	90-110		
Matrix Spike (BEL0595-MS4)			**Source: 23L0963-01**		Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	24.7		0.1	mg/L	5.000	19.5	102	90-110		
Reference (BEL0595-SRM1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0595-SRM2)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0595-SRM3)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0595-SRM4)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0595-SRM5)					Prepared & Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos, CA 93635

Account# 00-0022364
Account Manager: Ben Nydam
Submitted By: John & Nicole Santos

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0701										
Blank (BEL0701-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0701-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Blank (BEL0701-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	8.1		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0701-DUP1)		**Source: 23L0985-02**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.51		0.01	mmhos/cm		1.48			2.19	10
pH	7.7		1.0	units		7.7			0.130	10
Electrical Conductivity umhos	1510		10.0	umhos/cm		1480			2.19	10
Duplicate (BEL0701-DUP2)		**Source: 23L0989-03**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.84		0.01	mmhos/cm		1.83			0.730	10
Electrical Conductivity umhos	1840		10.0	umhos/cm		1830			0.730	10
pH	8.7		1.0	units		8.7			0.00	10
Reference (BEL0701-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	424			umhos/cm	426.0		99.6	90-110		
Reference (BEL0701-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0701-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0701-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0701-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		

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DELLAVALLE™
LABORATORY INC

John & Nicole Santos Dairy #2
22643 S Mercey Springs Rd
Los Banos, CA 93635

Account# 00-0022364
Account Manager: Ben Nydam
Submitted By: John & Nicole Santos

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0701 (Continued)										
Reference (BEL0701-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0701-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0701-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0701-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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WATER WORK REQUEST

Bill To: Acct No. 22364 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: **Santos Dairy, John & Nicole #2**
Address: 22643 S Mercy Springs Road
City, State, Zip: Los Banos, CA 93635
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: John & Nicole Santos - (209) 968-7832

Ranch _____

Date sampled: 12/13/23

Sampled by: M Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples 1 No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[Y] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(Y) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13	11:50	Ø	4.2
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeyrel	CAS INC	12/13/23 @ 11:50	12/14/23
Second	JLF	DLV	12-14-23 12:00	12-14-23 2:15
Third				
Fourth		DLV	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

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	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1									
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT